

Trunnion Mounted Ball Valve

Structural Features of Fixed Ball Valve

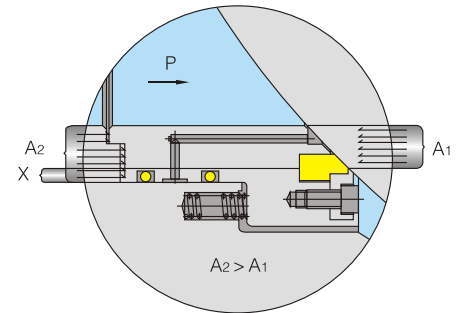
Fixed ball valves are mainly used in the industries of natural gas, oil products, chemicals, metallurgy, urban construction, environmental protection, pharmaceuticals, foodstuff and etc. Among them, the sulfur-resisting series products are applicable for seriously corrosive natural gas pipeline containing hydrogen sulphide mediums and impurities.

1. Distinctive Seating Structure

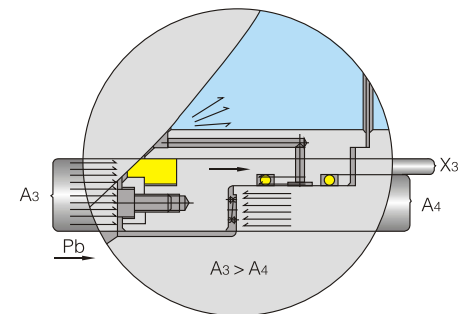
According to the extent of pressure, medium properties and sealing requirements, fixed ball valves may be made to front seal structure, back seal structure, or front-back dual seal structure.

● Front Seal Structure

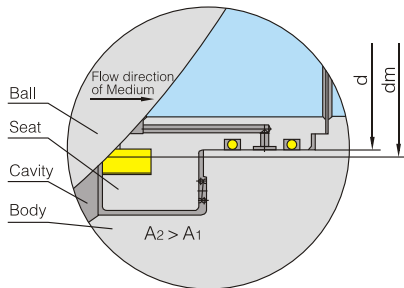
The front seal structure of valve seat gives the functions of two way seal and self-relieving function at middle cavity. As shown in the figure, the sealing socket inlaid with appropriate polymeric material (high molecular material of NYLON, MOLON, DELRIN or PEEK) is float. With spring loaded, when closed, the sealing face remains always in close contact with the ball, thus to ensure leak-tightness under whatever high or low differential pressure. Upstream: valve seat moves axially along with the valve, the upstream (inlet) pressure P applied to A_2 produces a directional force on A_1 . As A_2 is larger than A_1 , $A_2 - A_1 = X$, so the pressure on X will push the valve seat toward the ball to actualize close seal at upstream. Downstream: in case the pressure inside valve cavity P_b rises, the force acted upon A_3 will be greater than that upon A_4 , $A_3 - A_4 = X_2$. The differential pressure formed up on X_2 will overcome the spring force and make valve seat separated from ball, and then, the valve seat will be closed to the ball again under spring action.



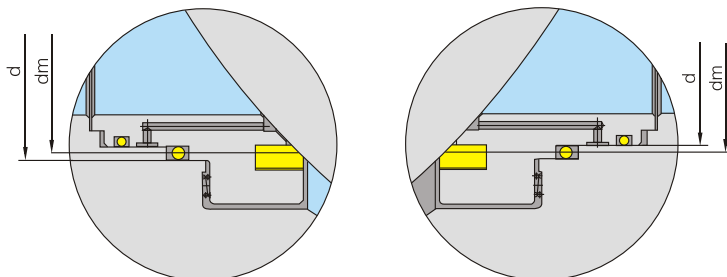
Inlet Side



Outlet Side



Outlet Side



Inlet Side

Outlet Side

Front-Back Seal Structure

● Back Seal Structure

The piston effect at valve seat produced by the area difference between 'd' and 'dm' (see figure on the right) will make seating seal ring and ball closely contacted and sealed under the medium action at the middle cavity of valve body.

● Front-back Seal Structure

At the inlet inside, the piston effect at valve seat produced by the area difference between 'd' and 'dm' will make seating seal ring and ball closely contacted and sealed under the medium action at upstream. At the outlet side, the piston effect produced by the area difference between 'd0' and 'dm' will make seating seal ring and ball closely contacted and sealed under the action of medium pressure at the middle cavity of valve body.

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2. Self-relieving Structure

In case of abnormal rise of pressure at middle cavity, ball valve of single seal structure is provided with self-relieving function, while ball valve of dual seal structure uses the auxiliary pressure relief device on valve body to carry out pressure relief.

3. First Aid of Seal

The valve is designed with an auxiliary seating emergency seal system, as shown in the figure. In case of soft seal damaged, or failure of seal in an emergency circumstances, emergency seal may be effected by injecting in sealant through auxiliary sealing system. If needed, emergency seal assembly can be used to rinse and lubricate the seating area. Plus, valve stem can also be designed with an auxiliary emergency sealing system.

4. Fire Protection Structure

If requested by working conditions and users, ball valve may be designed to fireproof. The fireproofing design of ball valve is to the standards of API607 and JB/T6899. In case of soft seal ring burnt in fire, the fire protection structure of ball valve functions to prevent mediums from mass leakage, thus to avoid fire spreading.

5. Antistatic Structure

When operating the valve, the friction between the ball and seat will produce electrostatic charge that can be accumulated on the ball. To prevent static spark, an antistatic device is placed on the valve to derive the electric charge accumulated on the ball (as shown in the figure).

6. Locking Device

To prevent misoperation and the unexpected open or close caused by the unpredicted circuit vibration, a locking device is designed at the fully opened and closed positions of hand operated ball valve. This design is proven especially good and effective in the production line of inflammable mediums of petroleum and chemicals, or when valves are mounted outdoors.

7. Full and Reduced Bore Structure

To meet the different requirements of users, we have full and reduced bore ball valves series (as shown in the figure). The inside diameter of full-bore ball valve is in conformity with that of the pipe for the convenience of cleaning, while reduced-bore series ball valve is comparatively lighter in weight, but its fluid resistance is only 1/7 of that of globe valve of the same caliber. So, reduced-bore ball valves boast of broader prospects.

8. Discharge Device on Valve Body

If requested by users or by the system, discharge valve may be mounted on the body of ball valve. In case the two ends of valve have been sealed, the pressure in valve will be released through the discharge valve on the body. Besides the function of DBB, this discharge valve also functions to rinse and blow out the deposits inside valve body.

9. Corrosion Resistance

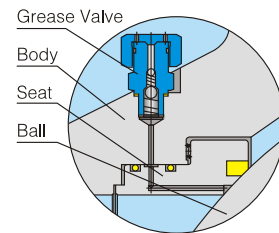
Corrosion allowance is left for the design of body thickness, the stem, fixed shaft, ball, seat and bottom cover of carbon steel valve are all chemical plated in surface to ASTM B733 and B656. Use of paint from international co. to deal with all kinds of conditions.

10. Sulfide Stress Cracking Resistance

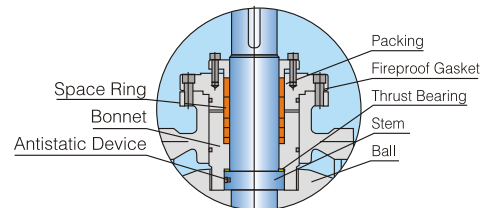
The materials exposed to fluid of our sulfur resisting ball valves, including fastening components, are all selected to NACE MR0175. Rigorous quality control and inspection is implemented in the manufacturing process to make our products conformed to the standard and suitable for vulcanizing conditions.

11. Extension Bar Device

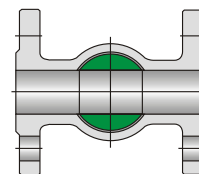
Extension device may be provided for buried ball valves, which include the extension of valve stem, greasing valve and discharge valve. As shown in the figure on the right, users shall specify the requirements and length to be extended (The length is generally the distance from the center of valve channel to the center of operating device).



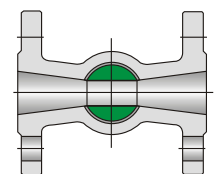
Seating Auxiliary Seal



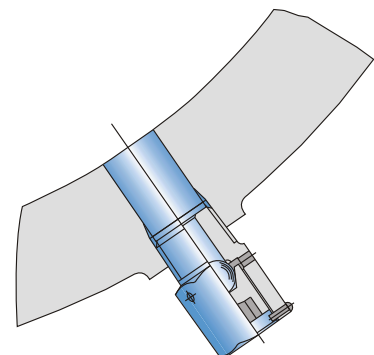
Stem Anti-flyout Structure
Stem Antistatic Device
Bonnet Leak-tight Structure



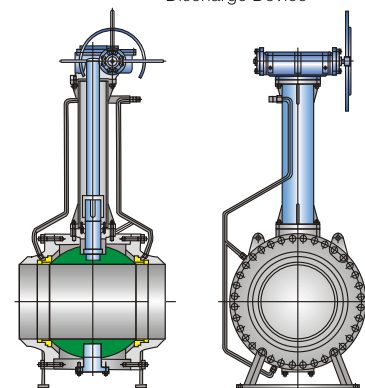
Full-bore Channel Type



Reduced-bore Channel Type



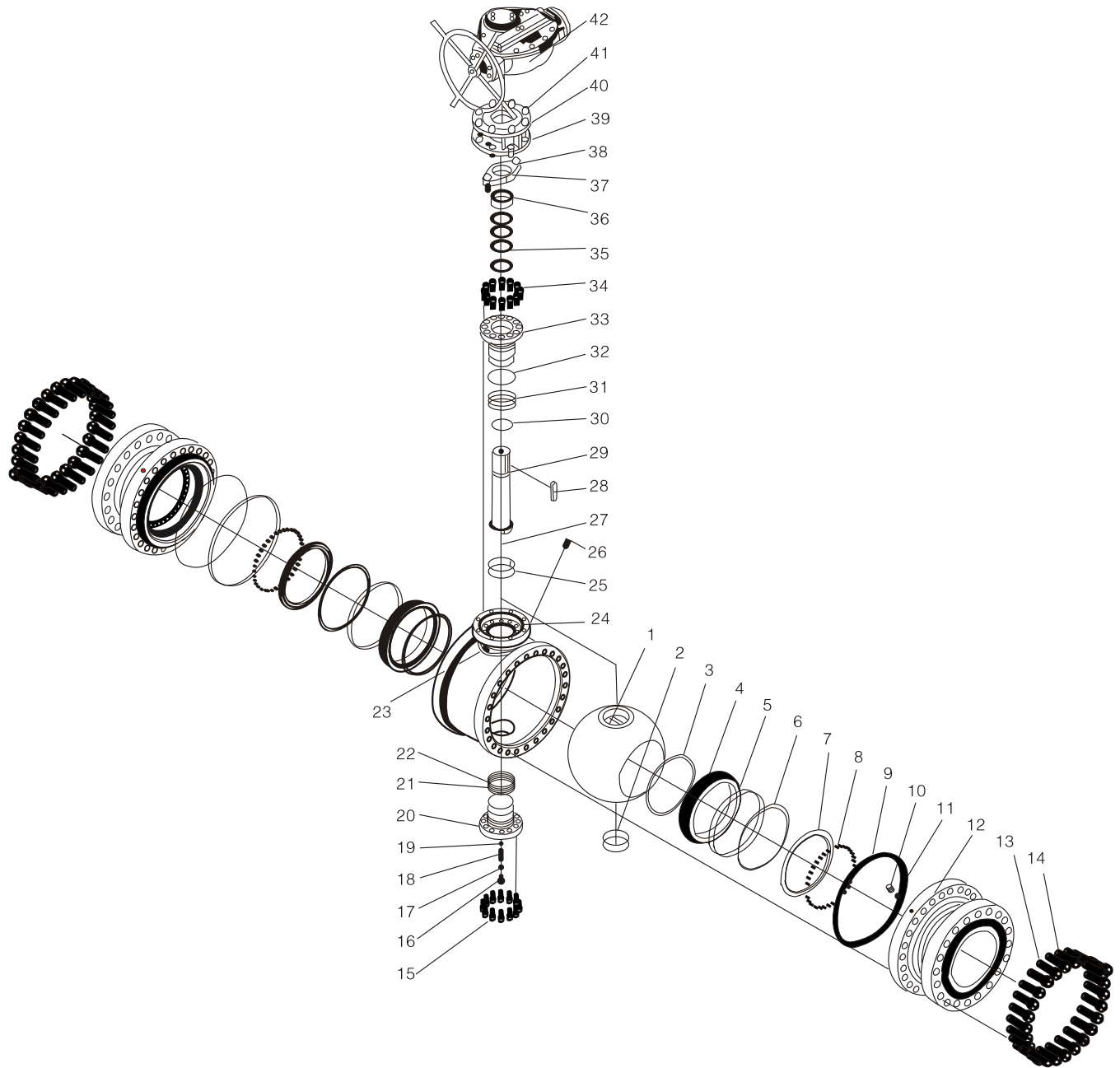
Discharge Device



Extension Bar Device

Trunnion Mounted Ball Valve

Design Drawing of Cast Steel Valve



Trunnion Mounted Ball Valve

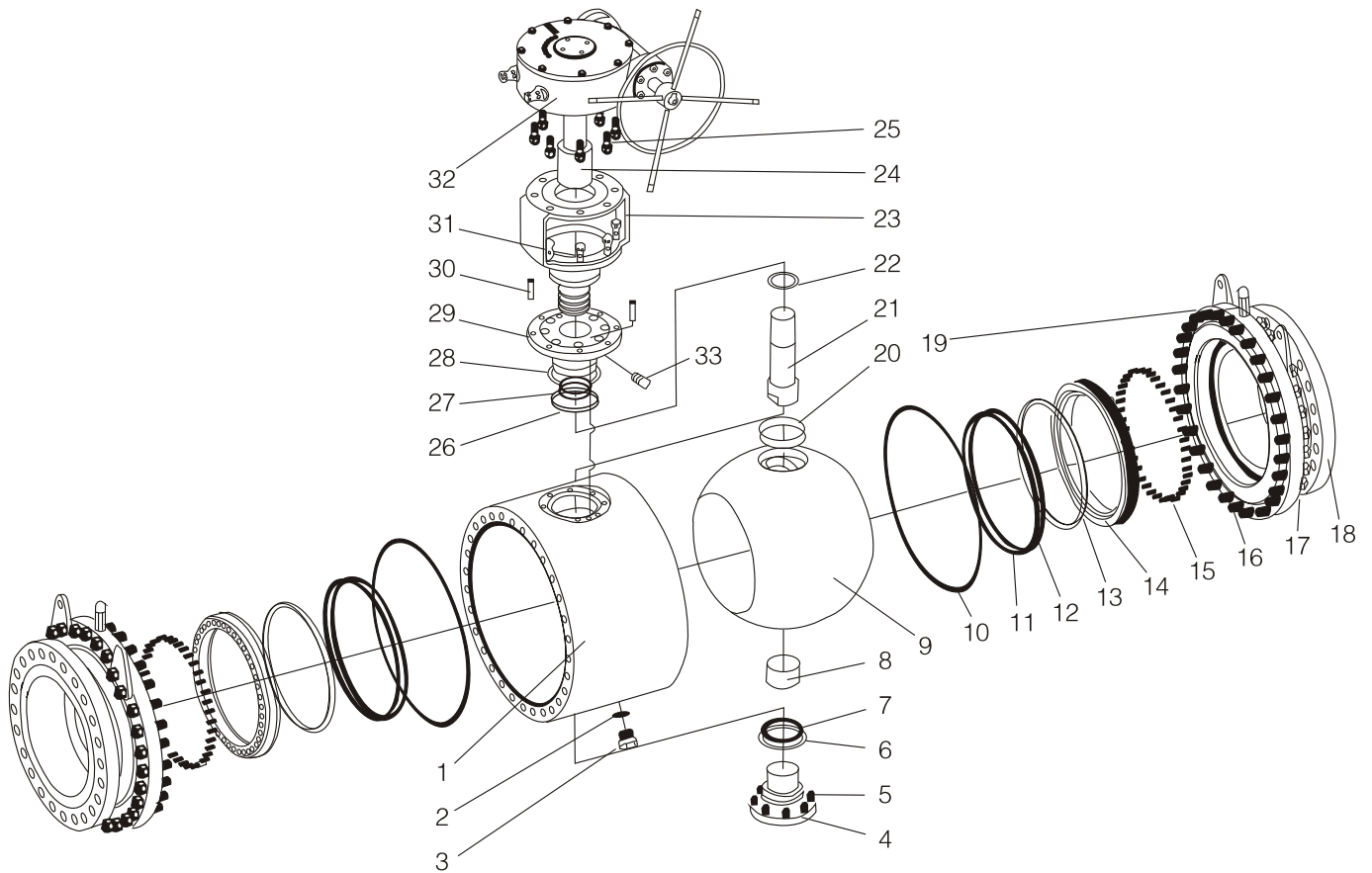
The Material of Main Parts

No.	Parts Name	CarbonSteel Series	Stainless steel Series	Low Temperature Series	Anti–Sulphur	
1	Ball	A105+HCr/ENP	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC+ENP	A105+HCr/ENP	A351 CF8M+ENP
2	Bush Bearing	Metal +PTFE,Braided Graphite Fibre				
3	Sealing Ring	PTFE、RPTFE、NYLON、MOLON、DELRIN or PEEK				
4	Seat	A105+HCr/ENP	A182、F304、316	A182、F6a	A105+HCr/ENP	A182、F316
5	O Ring	VITON				
6	Fire Safe Gasket	Soft Graphite+ SS				
7	Supporting Ring	A105+HCr/ENP	A182、F304、316	A182、F6a	A105+HCr/ENP	A182、F316
8	Spring			INCONEL 750		
9	O Ring		VITON			
10	Sealant Injection Valve	The same material with Body				
11	Gasket	Soft Graphite+ SS				
12	Bonnet	A216 WCB	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC	A216 WCB	A351 CF8M
13	Stud	193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
14	Nut	A194 2H	A194 8M	A194 4	A194 2HM	A194 2HM
15	Nipple	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
16	Plug	A105+HCr/ENP	A182、F304、316	A182 F6a	A105+HCr/ENP	A182 F316
17	Gasket	Gasket	RPTFE	RPTFE	Copper	RPTFE
18	Nipple	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
19	Ball	A182 F304				
20	Bottom cover	A105+HCr/ENP	A182、F304、316	A182 F6a	A105+A182 F316、HCr/ENP	
21	Gasket	Soft Graphite+ SS				
22	O Ring	VITON				
23	Sealant Injection Valve	The same material with Body				
24	Body	A216 WCB	A351、CF8、CF8M、CF3、CF3M	A352 LCB、LCC	GB/T12229、A216 WCB	A351 CF8M
25	Bush Bearing	Metal+PTFE,Braided Graphite Fibre				
26	Drainage Valves	The same material with Body				
27	Spring	A182 F304				
28	Key	GB/T699 45				
29	Stem	A182 F6a	A182 F304、316	A182 F6a	A182 F304	A182 F304
30	Thrust bearing	Metal+PTFE,Braided Graphite Fibre				
31	O Ring	VITON				
32	Gasket	Soft Graphite+ SS				
33	Back Seat	A105+HCr/ENP	A182 F304、316	A182 F6a	A105+HCr/ENP	A182 F316
34	Nipple	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
35	Packing	Soft Graphite, PTFE				
36	Packing Gland	A182 F6	A182 F304、316	A182 F304	A182 F6	A182 F316
37	Gland Flange	A216 WCB	A351 CF8、CF8M	A351 CF8	A216 WCB	A351 CF8M
38	Bolt	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
39	Bolt	A193 B7	A193 B8、B8M	A320 L7	A320 L7	A193 B7M
40	Yoke	A216 WCB	A351 CF8	A352 LCB	A216 WCB	A351 CF8
41	Bolt	A193 B7	A193 B8、B8M	A320 L7	A193 B7M	A193 B7M
42	Operation Type	Worm Gear,Electric Operation,Pneumatic Operation,Airdraulic Operation				

Note: The chart above only lists out some common composition of valve parts, We may provide other different parts material composition according to the customer's request or the actual valve working condition.

Trunnion Mounted Ball Valve

Valve Structural Drawing



Trunnion Mounted Ball Valve

The Material of Main Parts

No.	Parts Name	Material		
		Carbon Steel	SS	Low Temperature Steel
1	Body	ASTM A105	A182 F304	A350 LF2
2	Gasket	Soft Graphite+SS		
3	Drain Valve	ASTM A105	A182 F304	A350 LF2
4	Bottom Cover	ASTM A105	A182 F304	A350 LF2
5	Stud	A193 B7	A193 B8	A320 L7
6	Gasket	Soft Graphite+SS		
7	O Ring	VITON		
8	Bottom Bush Bearing	PTFE+CS	PTFE+SS	PTFE+SS
9	Ball	ASTM A105+ENP	A182 F304	A182 F304
10	Middle Flange Gasket	Soft Graphite+SS		
11	O Ring	VITON		
12	Fire Safe Ring	Soft Graphite		
13	Sealing Ring	PTFE、NYLON、PEEK、MOLON、PCTFE		
14	Seat	ASTM A105+ENP	A182 F304	A182 F304
15	Spring	INCONEL600		
16	Stud	A193 B7	A193 B8	A320 L7
17	Nut	A194 2H	A194 8	A194 7
18	Bonnet	ASTM A105	A182 F304	A350 LF2
19	Sealant Injection Valve	ASTM A105	A182 F304	A350 LF2
20	Top Bush Bearing	PTFE+CS	PTFE+SS	PTFE+SS
21	Stem	A182 F6a	A182 F304	A182 F304
22	Flat Bush Bearing	PTFE+CS	PTFE+SS	PTFE+SS
23	Yoke	A216 WCB		
24	Connecting Ring	ANSI 1025		
25	Stud	A193 B7		
26	O Ring	VITON		
27	O Ring	VITON		
28	Gasket	Soft Graphite+SS		
29	Back Seat	ASTM A105	A182 F304	A350 LF2
30	Locating Pin	A182 F6a		
31	Stud	A193 B7		
32	Worm Gear	-		
33	Sealant Injection Valve	ASTM A105	A182 F304	A350 LF2

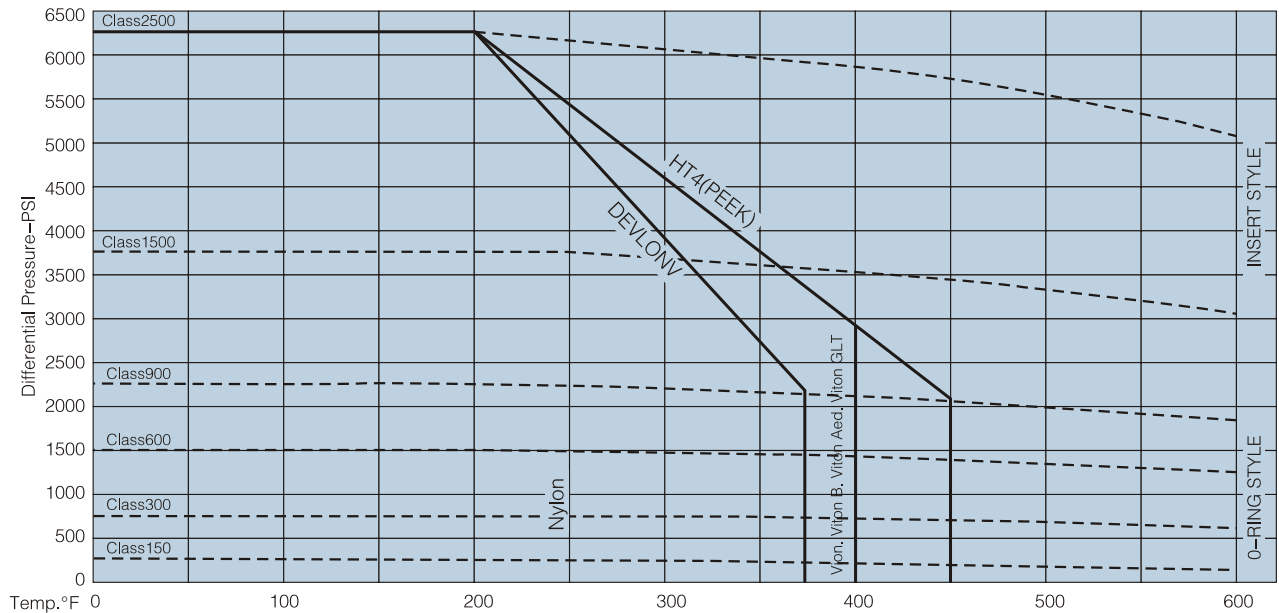
Note: The chart above only lists out some common composition of valve parts, We may provide other different parts material composition according to the customer's request or the actual valve working condition.

Trunnion Mounted Ball Valve

The specification of Sealing Seat Material

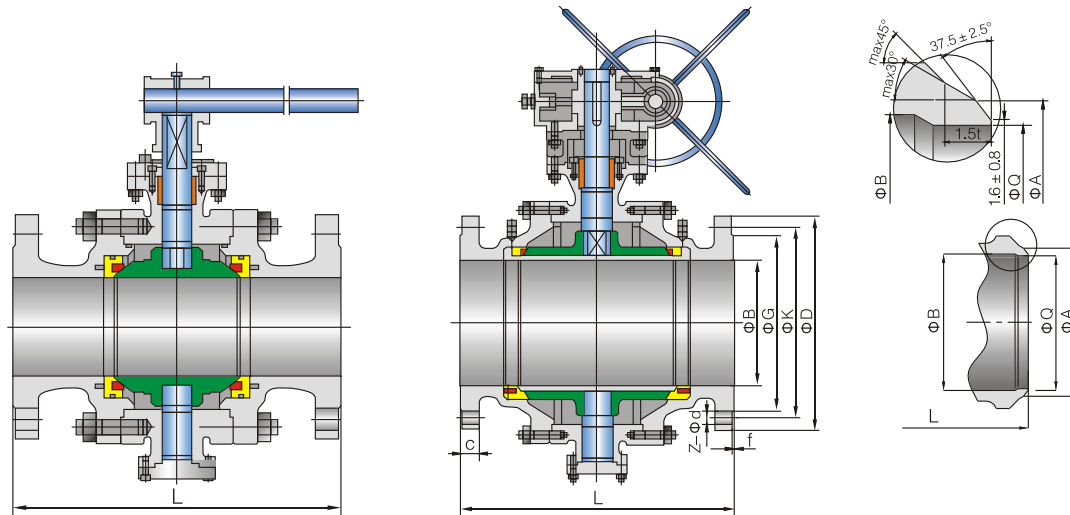
Test Standard	Test Item	Unit	PEEK	NOLON	DEVLON	PPL	PTFE	PTFE+ Graphite	PTFE+ Glass Fiber	NYLON66
D638	Tensile Strength 23°C/-°C	MPa	93.08	75/100	79.92/109.52	72	24.82	25	24.2	60/80
D638	Elongation at Break 23°C	%	50	10/30	5.37	6/8	300	150	105	60
D785	Hardness	D	–	78	78/80	80	56	58	65	78
		R	120	110/120	114	–	–	–	–	118
D790	Bend Strength	MPa	166.71	140	121.55	176	–	–	23.7	117
D621	Deforming under pressure during 24hours	%	~ 0	1.2	1.0/2.0	0.78	14/28	8.8	5.5	1.4
E831	Linear expansion factor	1/K	0.48×10^{-4}	0.6×10^{-4}	1.1×10^{-4}	0.43×10^{-4}	1.2×10^{-4}	1×10^{-4}	1×10^{-4}	0.7×10^{-4}
D648	1.82MPa/0.46MPa Heat Deforming Temperature	°C	160	150/190	93	163	55	63	78	90
		°C	–	–	209	–	132	–	–	235
D792	Density	G/cm ³	1.34 ~ 1.36	1.15	1.14	1.48	2.20	2.22	2.1	1.12
D570	24 hours Absorption rate	%	0.13	0.7	0.1	0.2	0.01	0.015	0.015	1.2
D695	Tensile Strength	MPa	142	140	140	117	35	45	52	–
D695	Yield Strength	MPa	–	120	88.9	–	11.7	–	–	75.8

Performance Data Sheet of Seat Sealing Materials



Above table gives the temperature and pressure ratings of nylon, devlonv, PEEK, viton and etc., and the temperature and pressure ratings of 150Lb, 300Lb and 600Lb equally apply to seat design with O-ring.

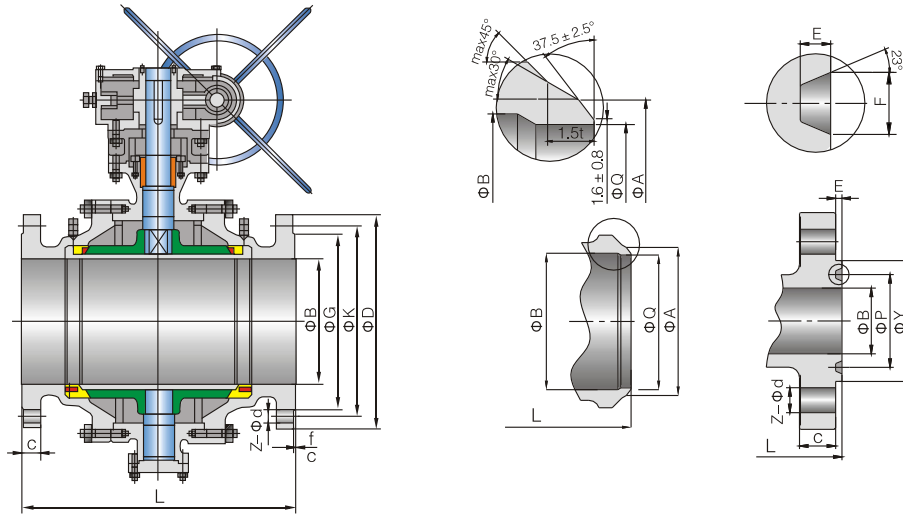
Full Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)										
	NPS	DN	L		B	D	G	f	K	Z-Φd	C	A	Q
			RF	BW									
150Lb	2	50	178	216	49	152	91.9	1.6	120.7	4-19.1	15.8	63	Asper client's requirement
	2 1/2	65	191	241	62	178	104.6	1.6	139.7	4-19.1	17.5	75	
	3	80	203	283	74	191	127	1.6	152.4	4-19.1	19.1	91	
	4	100	229	305	100	229	157.2	1.6	190.5	8-19.1	23.9	117	
	6	150	394	457	150	279	215.9	1.6	241.3	8-22.2	25.4	172	
	8	200	457	521	201	343	269.7	1.6	298.5	8-22.2	28.4	223	
	10	250	533	559	252	406	323.9	1.6	362	12-25.4	30.2	278	
	12	300	610	635	303	483	381	1.6	431.8	12-25.4	31.8	329	
	14	350	686	762	334	533	412.8	1.6	476.2	12-28.6	35.1	362	
	16	400	762	838	385	597	469.9	1.6	539.8	16-28.6	36.6	413	
	18	450	864	914	436	635	533.4	1.6	577.9	16-31.8	39.6	464	
	20	500	914	991	487	699	584.2	1.6	635	20-31.8	42.9	516	
	24	600	1067	1143	589	813	692.2	1.6	749.3	20-34.9	47.8	619	
	26	650	1143	1245	633	870	749.3	1.6	806.5	24-34.9	68.3	670	
	28	700	1245	1346	684	927	800.1	1.6	863.6	28-34.9	71.4	721	
	30	750	1295	1397	735	984	857.3	1.6	914.4	28-34.9	74.7	772	
	32	800	1372	1524	779	1060	914.4	1.6	977.9	28-41.3	80.8	825	
	34	850	1473	1626	830	1111	965.2	1.6	1028.7	32-41.3	82.6	876	
	36	900	1524	1727	874	1168	1022.4	1.6	1085.9	32-41.3	90.4	927	

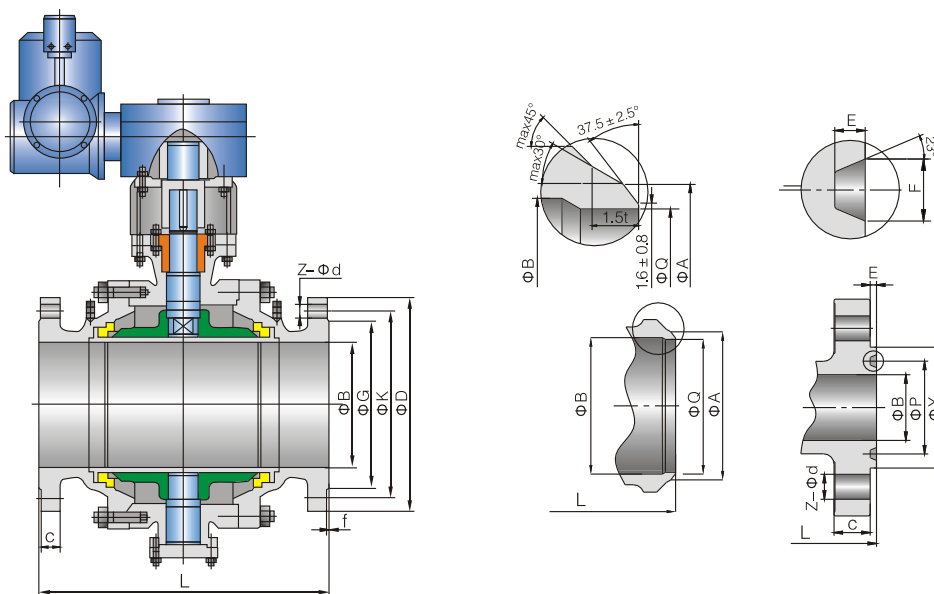
Full Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
300Lb	2	50	216	232	216	49	165	91.9	1.6	127	8-19.1	22.4	108	82.55	7.92	11.91	63	Asper client's requirement
	2 1/2	65	241	257	241	62	191	104.6	1.6	149.2	8-22.2	25.4	127	101.6	7.92	11.91	75	
	3	80	283	298	283	74	210	127	1.6	168.3	8-22.2	28.4	146	123.82	7.92	11.91	91	
	4	100	305	321	305	100	254	157.2	1.6	200.2	8-22.2	31.8	175	149.22	7.92	11.91	117	
	6	150	403	419	457	150	318	215.9	1.6	269.8	12-22.2	36.6	241	211.12	7.92	11.91	172	
	8	200	502	518	521	201	381	269.7	1.6	330.2	12-25.4	41.1	302	269.88	7.92	11.91	223	
	10	250	568	584	559	252	445	323.9	1.6	387.4	16-28.6	47.8	356	323.85	7.92	11.91	278	
	12	300	648	664	635	303	521	381	1.6	450.9	16-31.8	50.8	413	381	7.92	11.91	329	
	14	350	762	787	762	334	584	412.8	1.6	514.4	20-31.8	53.8	457	419.1	7.92	11.91	362	
	16	400	838	854	838	385	648	469.9	1.6	571.5	20-34.9	57.2	508	469.9	7.92	11.91	413	
	18	450	914	930	914	436	711	533.4	1.6	628.7	24-34.9	60.5	575	533.4	7.92	11.91	464	
	20	500	991	1010	991	487	775	584.2	1.6	685.8	24-34.9	63.5	635	584.2	9.52	13.49	516	
	24	600	1143	1165	1143	589	914	692.2	1.6	812.8	24-41.3	69.9	749	692.15	11.13	16.66	619	
	26	650	1245	1270	1245	633	972	749.3	1.6	876.3	28-44.5	79.2	809.8	749.3	12.7	19.84	670	
	28	700	1346	1372	1346	684	1035	800.1	1.6	939.8	28-44.5	85.9	860.6	800.1	12.7	19.84	721	
	30	750	1397	1422	1397	735	1092	857.3	1.6	997	28-47.6	91.9	917.4	857.25	12.7	19.84	772	
	32	800	1524	1533	1524	779	1149	914.4	1.6	1054.1	28-50.8	98.6	984.3	914.4	14.27	23.01	825	
	34	850	1626	1654	1626	830	1207	965.2	1.6	1104.9	28-50.8	101.6	1035.1	965.2	14.27	23.01	876	
	36	900	1727	1756	1727	874	1270	1022.4	1.6	1168.4	32-54	104.6	1092.2	1022.35	14.27	23.01	927	

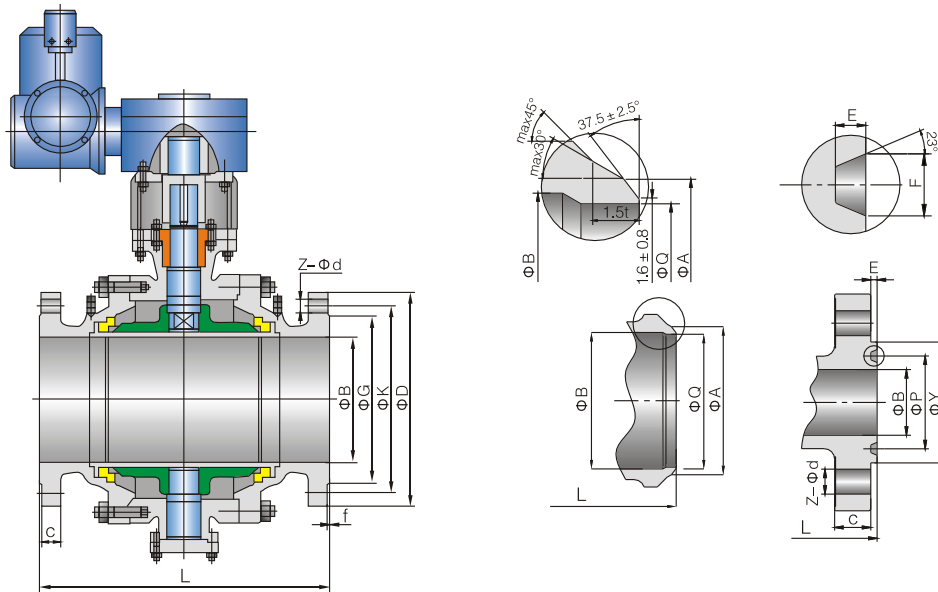
Full Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
600Lb	2	50	292	295	292	49	165	91.9	6.4	127	8-19.1	25.4	108	82.55	7.92	11.91	63	Asper client's requirement
	2 1/2	65	330	333	330	62	191	104.6	6.4	149.2	8-22.2	28.4	127	101.6	7.92	11.91	75	
	3	80	356	359	356	74	210	127	6.4	168.3	8-22.2	31.8	146	123.82	7.92	11.91	91	
	4	100	432	435	432	100	273	157.2	6.4	215.9	8-25.4	38.1	175	149.22	7.92	11.91	117	
	6	150	559	562	559	150	356	215.9	6.4	292.1	12-28.6	47.8	241	211.12	7.92	11.91	172	
	8	200	660	664	660	201	419	269.7	6.4	349.2	12-31.8	55.6	302	269.88	7.92	11.91	223	
	10	250	787	791	787	252	508	323.9	6.4	431.8	16-34.9	63.5	356	323.85	7.92	11.91	278	
	12	300	838	841	838	303	559	381	6.4	489	20-34.9	66.5	413	381	7.92	11.91	329	
	14	350	889	892	889	334	603	412.8	6.4	527.1	20-38.1	69.9	457	419.1	7.92	11.91	362	
	16	400	991	994	991	385	686	469.9	6.4	603.2	20-41.3	76.2	508	469.9	7.92	11.91	413	
	18	450	1092	1095	1092	436	743	533.4	6.4	654.1	20-44.5	82.6	575	533.4	7.92	11.91	464	
	20	500	1194	1200	1194	487	813	584.2	6.4	723.9	24-44.5	88.9	635	584.2	9.52	13.49	516	
	24	600	1397	1407	1397	589	940	692.2	6.4	838.2	24-50.8	101.6	749	692.15	11.13	16.66	619	
	26	650	1448	1461	1448	633	1016	749.3	6.4	914.4	28-50.8	108	809.8	749.3	12.7	19.84	670	
	28	700	1549	1562	1549	684	1073	800.1	6.4	965.2	28-54	111.3	860.6	800.1	12.7	19.84	721	
	30	750	1651	1664	1651	735	1130	857.3	6.4	1022.4	28-54	114.3	917.4	857.25	12.7	19.84	772	
	32	800	1778	1794	1778	779	1194	914.4	6.4	1079.5	28-60.3	117.3	984.3	914.4	14.27	23.01	825	
	34	850	1930	1946	1930	830	1245	965.2	6.4	1130.3	28-60.3	120.7	1035.1	965.2	14.27	23.01	876	
	36	900	2083	2099	2083	874	1314	1022.4	6.4	1193.8	28-66.7	124	1092.2	1022.35	14.27	23.01	927	

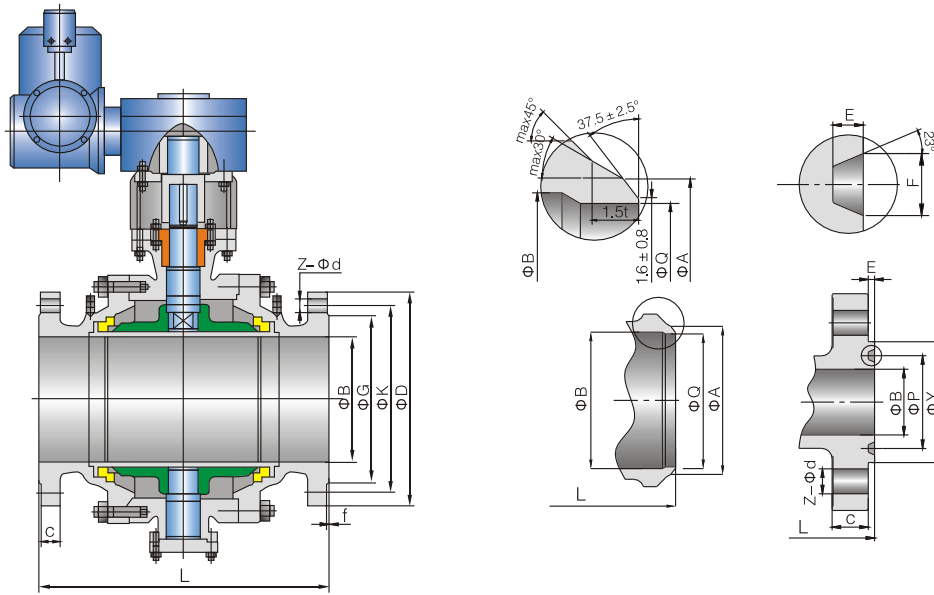
Full Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)															Q
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	
			RF	RTJ	BW								Y	P	E	F		
900Lb	2	50	368	371	368	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91	63	Asper client's requirement
	2 1/2	65	419	422	419	62	244	104.6	6.4	190.5	8-28.6	41.1	137	107.95	7.92	11.91	75	
	3	80	381	384	381	74	241	127	6.4	190.5	8-25.4	38.1	156	123.82	7.92	11.91	91	
	4	100	457	460	457	100	292	157.2	6.4	235	8-31.8	44.5	181	149.22	7.92	11.91	117	
	6	150	610	613	610	150	381	215.9	6.4	317.5	12-31.8	55.6	241	211.12	7.92	11.91	172	
	8	200	737	740	737	201	470	269.7	6.4	393.7	12-38.1	63.5	308	269.88	7.92	11.91	223	
	10	250	838	841	838	252	546	323.9	6.4	469.9	16-38.1	69.9	362	323.85	7.92	11.91	278	
	12	300	965	968	965	303	610	381	6.4	533.4	20-38.1	79.2	419	381	7.92	11.91	329	
	14	350	1029	1038	1029	322	641	412.8	6.4	558.8	20-41.3	85.9	467	419.1	11.13	16.66	362	
	16	400	1130	1140	1130	373	705	469.9	6.4	616	20-44.5	88.9	524	469.9	11.13	16.66	413	
	18	450	1219	1232	1219	423	787	533.4	6.4	685.8	20-50.8	101.6	594	533.4	12.7	19.84	464	
	20	500	1321	1334	1321	471	857	584.2	6.4	749.3	20-54	108	648	584.2	12.7	19.84	516	
	24	600	1549	1568	1549	570	1041	692.2	6.4	901.7	20-66.7	139.7	772	692.15	15.88	26.97	619	

Full Bore Trunnion Mounted Ball Valve

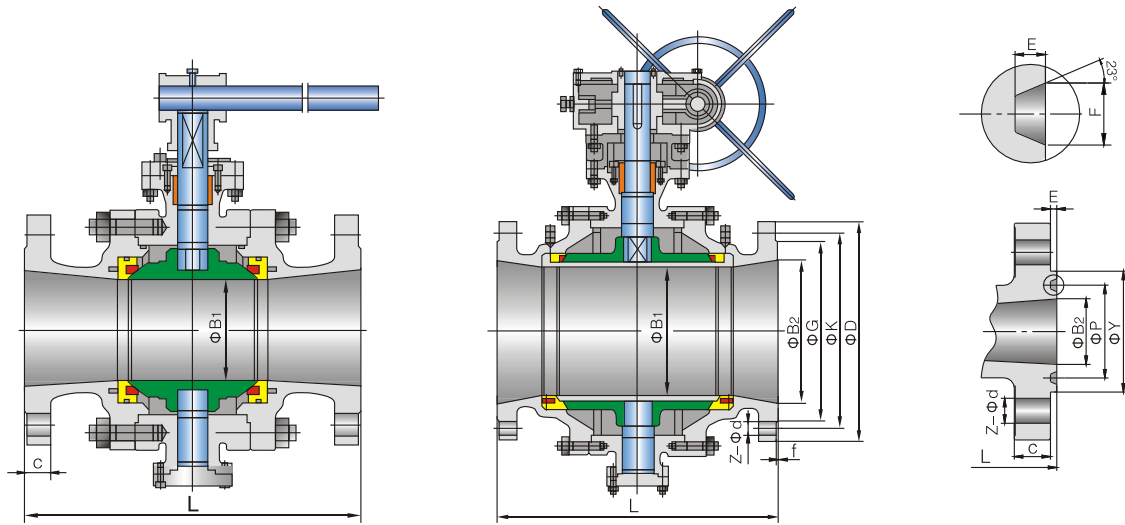


Main Connection Dimension

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
1500Lb	2	50	368	371	368	49	216	91.9	6.4	165.1	8-25.4	38.1	124	95.25	7.92	11.91	63	Asper client's requirement
	2 1/2	65	419	422	419	62	244	104.6	6.4	190.5	8-28.6	41.1	137	107.95	7.92	11.91	75	
	3	80	470	473	470	74	267	127	6.4	203.2	8-31.8	47.8	168	136.52	7.92	11.91	91	
	4	100	546	549	546	100	311	157.2	6.4	241.3	8-34.9	53.8	194	161.92	7.92	11.91	117	
	6	150	705	711	705	144	394	215.9	6.4	317.5	12-38.1	82.6	248	211.12	9.52	13.49	172	
	8	200	832	841	832	192	483	269.7	6.4	393.7	12-44.5	91.9	318	269.88	11.13	16.66	223	
	10	250	991	1000	991	239	584	323.9	6.4	482.6	12-50.8	108	371	323.85	11.13	16.66	278	
	12	300	1130	1146	1130	287	673	381	6.4	571.5	16-54	124	438	381	14.27	23.01	329	
	14	350	1257	1276	1257	315	749	412.8	6.4	635	16-60.3	133.4	489	419.1	15.88	26.97	362	
	16	400	1384	1407	1384	360	826	469.9	6.4	704.9	16-66.7	146.1	546	469.9	17.48	30.18	413	
	18	450	1537	1559	1537	406	914	533.4	6.4	774.7	16-73	162.1	613	533.4	17.48	30.18	464	
	20	500	1664	1686	1664	454	984	584.2	6.4	831.9	16-79.4	177.8	673	584.2	17.48	33.32	516	
	24	600	1943	1972	1943	546	1168	692.2	6.4	990.6	16-92	203.2	794	692.15	20.62	36.53	619	

Class	Size		Dimensions(mm)															
	NPS	DN	L			B	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW								Y	P	E	F		
2500Lb	2	50	451	454	451	42	235	91.9	6.4	171.5	8-28.6	50.8	133	101.6	7.92	11.91	63	Asper client's requirement
	2 1/2	65	508	514	508	52	267	104.6	6.4	196.9	8-31.8	57.2	149	111.12	9.52	13.49	75	
	3	80	578	584	578	62	305	127	6.4	228.6	8-34.9	66.5	168	127	9.52	13.49	91	
	4	100	673	683	673	87	356	157.2	6.4	273.1	8-41.3	76.2	203	157.18	11.13	16.66	117	
	6	150	914	927	914	131	483	215.9	6.4	368.3	8-54	108	279	228.6	12.7	19.84	172	
	8	200	1022	1038	1022	179	552	269.7	6.4	438.2	12-54	127	340	279.4	14.27	23.01	223	
	10	250	1270	1292	1270	223	673	323.9	6.4	539.7	12-66.7	165.1	425	342.9	17.48	30.18	278	
	12	300	1422	1445	1422	265	762	381	6.4	619.2	12-73	184.2	495	406.4	17.48	33.32	329	

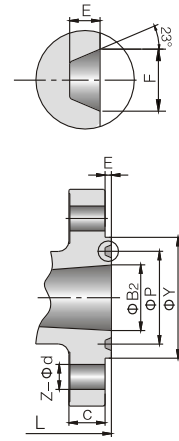
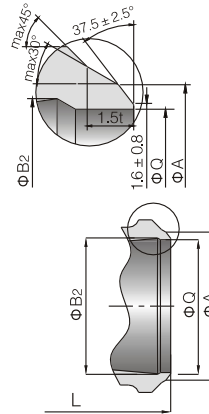
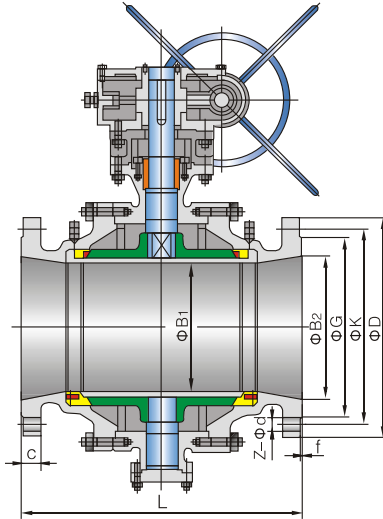
Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)											
	NPS	DN	L		B ₁	B ₂	D	G	f	K	Z-Φd	C	A	Q
			RF	BW										
150Lb	3 × 2	80 × 50	203	283	49	74	191	127	1.6	152.4	4-19.1	19.1	91	Asper client's requirement
	4 × 3	100 × 80	229	305	74	100	229	157.2	1.6	190.5	8-19.1	23.9	117	
	6 × 4	150 × 100	394	457	100	150	279	215.9	1.6	241.3	8-22.2	25.4	172	
	8 × 6	200 × 150	457	521	150	201	343	269.7	1.6	298.5	8-22.2	28.4	223	
	10 × 8	250 × 200	533	559	201	252	406	323.9	1.6	362	12-25.4	30.2	278	
	12 × 10	300 × 250	610	635	252	303	483	381	1.6	431.8	12-25.4	31.8	329	
	14 × 12	350 × 300	686	762	303	334	533	412.8	1.6	476.2	12-28.6	35.1	362	
	16 × 14	400 × 350	762	838	334	385	597	469.9	1.6	539.8	16-28.6	36.6	413	
	18 × 16	450 × 400	864	914	385	436	635	533.4	1.6	577.9	16-31.8	39.6	464	
	20 × 18	500 × 450	914	991	436	487	699	584.2	1.6	635	20-31.8	42.9	516	
	22 × 20	550 × 500	991	1092	487	538	750	641	1.6	692	20-34.9	46.0	567	
	24 × 22	600 × 550	1067	1143	538	589	813	692.2	1.6	749.3	20-34.9	47.8	619	
	26 × 24	650 × 600	1143	1245	589	633	870	749.3	1.6	806.5	24-34.9	68.3	670	
	28 × 26	700 × 650	1245	1346	633	684	927	800.1	1.6	863.6	28-34.9	71.4	721	
	30 × 28	750 × 700	1295	1397	684	735	984	857.3	1.6	914.4	28-34.9	74.7	772	
	32 × 30	800 × 750	1372	1524	735	779	1060	914.4	1.6	977.9	28-41.3	80.8	825	
	34 × 32	850 × 800	1473	1626	779	830	1111	965.2	1.6	1028.7	32-41.3	82.6	876	
	36 × 34	900 × 850	1524	1727	830	874	1168	1022.4	1.6	1085.9	32-41.3	90.4	927	

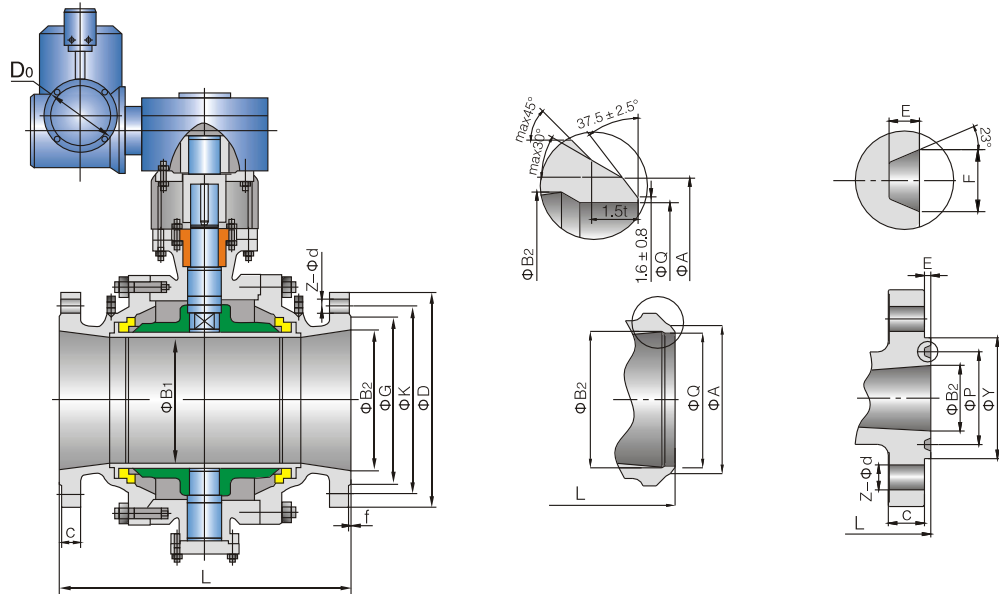
Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)																A	Q
	NPS	DN	L			B ₁	B ₂	D	G	f	K	Z-Φd	C	Groove						
			RF	RTJ	BW									Y	P	E	F			
300Lb	3 × 2	80 × 50	283	298	283	49	74	210	127	1.6	168.3	8-22.2	28.4	146	123.82	7.92	11.91	91	Asper client's requirement	
	4 × 3	100 × 80	305	321	305	74	100	254	157.2	1.6	200.2	8-22.2	31.8	175	149.22	7.92	11.91	117		
	6 × 4	150 × 100	403	419	403	100	150	318	215.9	1.6	269.8	12-22.2	36.6	241	211.12	7.92	11.91	172		
	8 × 6	200 × 150	502	518	521	150	201	381	269.7	1.6	330.2	12-25.4	41.1	302	269.88	7.92	11.91	223		
	10 × 8	250 × 200	568	584	559	201	252	445	323.9	1.6	387.4	16-28.6	47.8	356	323.85	7.92	11.91	278		
	12 × 10	300 × 250	648	664	635	252	303	521	381	1.6	450.9	16-31.8	50.8	413	381	7.92	11.91	329		
	14 × 12	350 × 300	762	787	762	303	334	584	412.8	1.6	514.4	20-31.8	53.8	457	419.1	7.92	11.91	362		
	16 × 14	400 × 350	838	854	838	334	385	648	469.9	1.6	571.5	20-34.9	57.2	508	469.9	7.92	11.91	413		
	18 × 16	450 × 400	914	930	914	385	436	711	533.4	1.6	628.7	24-34.9	60.5	575	533.4	7.92	11.91	464		
	20 × 18	500 × 450	991	1010	991	436	487	775	584.2	1.6	685.8	24-34.9	63.5	635	584.2	9.52	13.49	516		
	22 × 20	550 × 500	1092	1114	1092	487	538	840	641	1.6	743.0	24-41.3	66.5	686	635	11.13	15.09	567		
	24 × 22	600 × 550	1143	1165	1143	538	589	914	692.2	1.6	812.8	24-41.3	69.9	749	692.15	11.13	16.66	619		
	26 × 24	650 × 600	1245	1270	1245	589	633	972	749.3	1.6	876.3	28-44.5	79.2	809.8	749.3	12.7	19.84	670		
	28 × 26	700 × 650	1346	1372	1346	633	684	1035	800.1	1.6	939.8	28-44.5	85.9	860.6	800.1	12.7	19.84	721		
	30 × 28	750 × 700	1397	1422	1397	684	735	1092	857.3	1.6	997	28-47.6	91.9	917.4	857.25	12.7	19.84	772		
	32 × 30	800 × 750	1524	1533	1524	735	779	1149	914.4	1.6	1054.1	28-50.8	98.6	984.3	914.4	14.27	23.01	825		
	34 × 32	850 × 800	1626	1654	1626	779	830	1207	965.2	1.6	1104.9	28-50.8	101.6	1035.1	965.2	14.27	23.01	876		
	36 × 34	900 × 850	1727	1756	1727	830	874	1270	1022.4	1.6	1168.4	32-54	104.6	1092.2	1022.35	14.27	23.01	927		

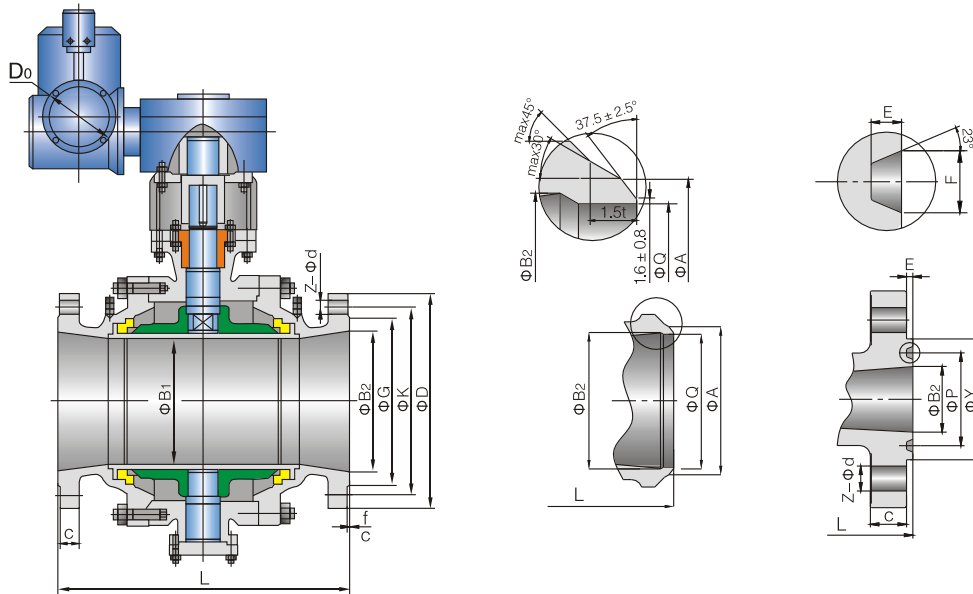
Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)																
	NPS	DN	L			B ₁	B ₂	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW									Y	P	E	F		
600Lb	3 × 2	80 × 50	356	359	356	49	74	210	127	6.4	168.3	8-22.2	31.8	146	123.82	7.92	11.91	91	Asper client's requirement
	4 × 3	100 × 80	432	435	432	74	100	273	157.2	6.4	215.9	8-25.4	38.1	175	149.22	7.92	11.91	117	
	6 × 4	150 × 100	559	562	559	100	150	356	215.9	6.4	292.1	12-28.6	47.8	241	211.12	7.92	11.91	172	
	8 × 6	200 × 150	660	664	660	150	201	419	269.7	6.4	349.2	12-31.8	55.6	302	269.88	7.92	11.91	223	
	10 × 8	250 × 200	787	791	787	201	252	508	323.9	6.4	431.8	16-34.9	63.5	356	323.85	7.92	11.91	278	
	12 × 10	300 × 250	838	841	838	252	303	559	381	6.4	489	20-34.9	66.5	413	381	7.92	11.91	329	
	14 × 12	350 × 300	889	892	889	303	334	603	412.8	6.4	527.1	20-38.1	69.9	457	419.1	7.92	11.91	362	
	16 × 14	400 × 350	991	994	991	334	385	686	469.9	6.4	603.2	20-41.3	76.2	508	469.9	7.92	11.91	413	
	18 × 16	450 × 400	1092	1095	1092	385	436	743	533.4	6.4	654.1	20-44.5	82.6	575	533.4	7.92	11.91	464	
	20 × 18	500 × 450	1194	1200	1194	436	487	813	584.2	6.4	723.9	24-44.5	88.9	635	584.2	9.52	13.49	516	
	22 × 20	550 × 500	1295	1305	1295	487	538	870	641	6.4	778	24-47.6	95	686	635.0	11.13	15.09	567	
	24 × 22	600 × 550	1397	1407	1397	538	589	940	692.2	6.4	838.2	24-50.8	101.6	749	692.15	11.13	16.66	619	
	26 × 24	650 × 600	1448	1461	1448	589	633	1016	749.3	6.4	914.4	28-50.8	108	809.8	749.3	12.7	19.84	670	
	28 × 26	700 × 650	1549	1562	1549	633	684	1073	800.1	6.4	965.2	28-54	111.3	860.6	800.1	12.7	19.84	721	
	30 × 28	750 × 700	1651	1664	1651	684	735	1130	857.3	6.4	1022.4	28-54	114.3	917.4	857.25	12.7	19.84	772	
	32 × 30	800 × 750	1778	1794	1778	735	779	1194	914.4	6.4	1079.5	28-60.3	117.3	984.3	914.4	14.27	23.01	825	
	34 × 32	850 × 800	1930	1946	1930	779	830	1245	965.2	6.4	1130.3	28-60.3	120.7	1035.1	965.2	14.27	23.01	876	
36 × 34	900 × 850	2083	2099	2083	830	874	1314	1022.4	6.4	1193.8	28.66.7	124	1092.2	1022.35	14.27	23.01	927		

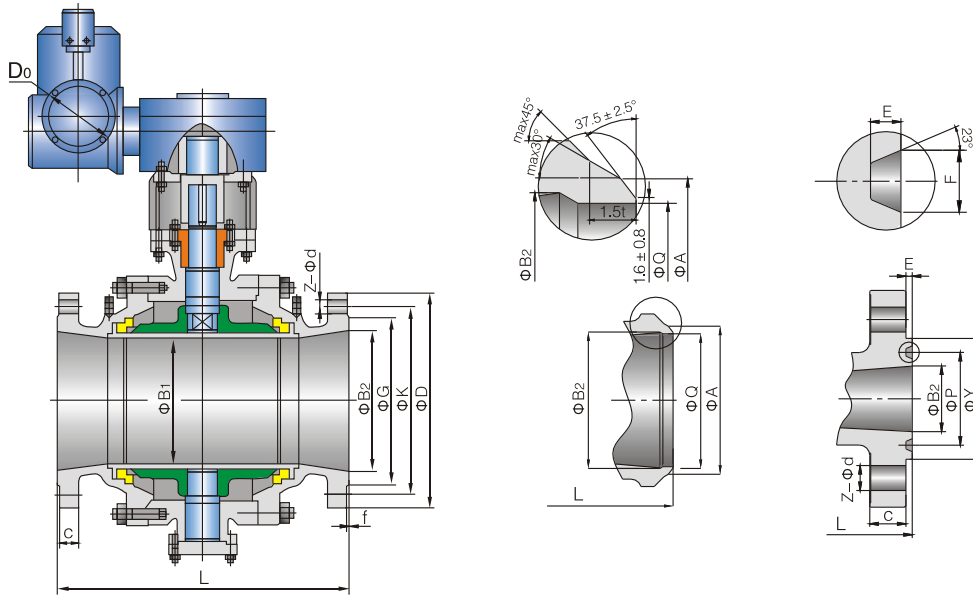
Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)																
	NPS	DN	L			B ₁	B ₂	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW									Y	P	E	F		
900Lb	3 × 2	80 × 50	381	384	381	49	74	241	127	6.4	190.5	8–25.4	38.1	156	123.82	7.92	11.91	91	Asper client's requirement
	4 × 3	100 × 80	457	460	457	74	100	292	157.2	6.4	235	8–31.8	44.5	181	149.22	7.92	11.91	117	
	6 × 4	150 × 100	610	613	610	100	150	381	215.9	6.4	317.5	12–31.8	55.6	241	211.12	7.92	11.91	172	
	8 × 6	200 × 150	737	740	737	150	201	470	269.7	6.4	393.7	12–38.1	63.5	308	269.88	7.92	11.91	223	
	10 × 8	250 × 200	838	841	838	201	252	546	323.9	6.4	469.9	16–38.1	69.9	362	323.85	7.92	11.91	278	
	12 × 10	300 × 250	965	968	965	252	303	610	381	6.4	533.4	20–38.1	79.2	419	381	7.92	11.91	329	
	14 × 12	350 × 300	1029	1038	1029	303	322	641	412.8	6.4	558.8	20–41.3	85.9	467	419.1	11.13	16.66	362	
	16 × 14	400 × 350	1130	1140	1130	322	373	705	469.9	6.4	616	20–44.5	88.9	524	469.9	11.13	16.66	413	
	18 × 16	450 × 400	1219	1232	1219	373	423	787	533.4	6.4	685.8	20–50.8	101.6	594	533.4	12.7	19.84	464	
	20 × 18	500 × 450	1321	1334	1321	423	471	857	584.2	6.4	749.3	20.54	108.0	648	584.2	12.7	19.84	516	

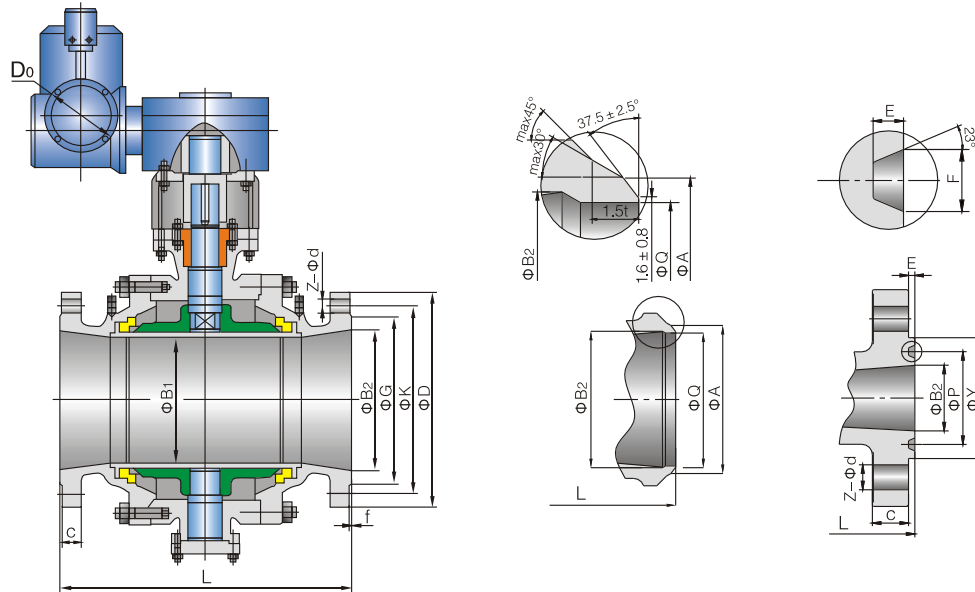
Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)																
	NPS	DN	L			B ₁	B ₂	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW									Y	P	E	F		
1500Lb	3×2	80×50	470	473	470	49	74	267	127	6.4	203.2	8-31.8	47.8	168	136.52	7.92	11.91	91	Asper client's requirement
	4×3	100×80	546	549	546	74	100	311	157.2	6.4	241.3	8-34.9	53.8	194	161.92	7.92	11.91	117	
	6×4	150×100	705	711	705	100	144	394	215.9	6.4	317.5	12-38.1	82.6	248	211.12	9.52	13.49	172	
	8×6	200×150	832	841	832	144	192	483	269.7	6.4	393.7	12-44.5	91.9	318	269.88	11.13	16.66	223	
	10×8	250×200	991	1000	991	192	239	584	323.9	6.4	482.6	12-50.8	108	371	323.85	11.13	16.66	278	
	12×10	300×250	1130	1146	1130	239	287	673	381	6.4	571.5	16-54	124	438	381	14.27	23.01	329	
	14×12	350×300	1257	1276	1257	287	315	749	412.8	6.4	635	16-60.3	133.4	489	419.1	15.88	26.97	362	
	16×14	400×350	1384	1407	1384	315	360	826	469.9	6.4	704.9	16-66.7	146.1	546	469.9	17.48	30.18	413	
	18×16	450×400	1537	1559	1537	360	406	914	533.4	6.4	774.7	16-73	162.1	613	533.4	17.48	30.18	464	
	20×18	500×450	1664	1686	1664	406	454	984	584.2	6.4	831.9	16-79.4	177.8	673	584.2	17.48	33.32	516	

Reduced Bore Trunnion Mounted Ball Valve



Main Connection Dimension

Class	Size		Dimensions(mm)																
	NPS	DN	L			B ₁	B ₂	D	G	f	K	Z-Φd	C	Groove				A	Q
			RF	RTJ	BW									Y	P	E	F		
2500Lb	3 × 2	80 × 50	578	584	578	42	62	305	127	6.4	228.6	8-34.9	66.5	168	127	9.52	13.49	91	Asper client's requirement
	4 × 3	100 × 80	673	683	673	62	87	356	157.2	6.4	273.1	8-41.3	76.2	203	157.18	11.13	16.66	117	
	6 × 4	150 × 100	914	927	914	87	131	483	215.9	6.4	368.3	8-54	108	279	228.6	12.7	19.84	172	
	8 × 6	200 × 150	1022	1038	1022	131	179	552	269.7	6.4	438.2	12-54	127	340	279.4	14.27	23.01	223	
	10 × 8	250 × 200	1270	1292	1270	179	223	673	323.9	6.4	539.7	12-66.7	165.1	425	342.9	17.48	30.18	278	
	12 × 10	300 × 250	1422	1445	1422	223	265	762	381	6.4	619.2	12-73	184.2	495	406.4	17.48	33.32	329	